| | |
|---|---|
| United States Patent Application Publication | 20250276781 |
| Kind Code | A1 |
| Publication Date | September 04, 2025 |
| Inventor(s) | Cretnik; Gregor et al. |


## Wing-Fuselage Truss Joint


### Abstract

A wing-fuselage truss joint assembly for an aircraft includes joining members secured to the aircraft's wing and fuselage structures. The joining members each include forward and aft wing joints and forward and aft fuselage joints connected by an upright member. The forward joining members are pivotally connected using interconnecting members and the aft joining members are connected using aft truss members. The forward-right and aft-right joining members and the forward-left and aft-left joining members are connected by truss link members and diagonal truss link members. The wing-fuselage truss joint assembly is configured to reduce forces experienced by the wing from being transferred to the aircraft fuselage.



**Inventors:** **Cretnik; Gregor (Ljubljana, SI), Marinkovic; Filip (Ajdovšcina, SI), Toš; Patrik (Puce, SI)**

**Applicant:** **Pipistrel d.o.o.** (Ajdovšcina, SI)



**Family ID:** **94820780**

**Appl. No.:** **18/591363**

**Filed:** **February 29, 2024**

### Publication Classification

**Int. Cl.:** **B64C1/26** (20060101); **B64C3/18** (20060101); **B64C3/38** (20060101)

**U.S. Cl.:**

CPC **B64C1/26** (20130101); **B64C3/185** (20130101); **B64C3/385** (20130101);


### Background/Summary

## BACKGROUND OF THE INVENTION

### 1. Field

[0001] The disclosed embodiments relate generally to the field of aircraft. More specifically, the disclosed embodiments relate to the field of aircraft wing and fuselage connections.

### 2. Description of the Related Art

[0002] It is known to use pin joints to attach a wing to a fuselage on an aircraft. For example, in U.S. Pat. No. 7,887,009 to Keeler et al. the wing is attached to the fuselage using pin joints designed to transfer loads from the wings to the fuselage.

[0003] It is also known for a system to mount pylon assemblies in a tiltrotor aircraft. For example, in U.S. Pat. No. 10,040,562 to Kooiman et al. and U.S. Pat. No. 9,981,750 to Williams et al. describes similar systems using spherical bearings to secure pylons to a fuselage or for mounting other major aircraft components to an aircraft.

## SUMMARY

[0004] This summary is provided to introduce a selection of concepts in a simplified form that are further described below in the detailed description. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. Other aspects and advantages will be apparent from the following detailed description of the embodiments and the accompanying drawing figures.

[0005] In some embodiments, the techniques described herein relate to a wing-fuselage truss joint assembly for an aircraft including a forward-left joining member, an aft-left joining member, a forward-right joining member, and an aft-right joining member configured to join a wing to a fuselage. Each of the joining members includes a wing truss joint mechanically coupled to a spar of the wing; a fuselage truss joint mechanically coupled to a bulkhead of the fuselage; and an upright member pivotally coupled to the wing truss joint at a first end and pivotally coupled to the fuselage truss joint at a second end; an interconnecting joint mechanically coupled to a forward bulkhead between the forward-left joining member and the forward-right joining member; at least one interconnecting member pivotally coupled to the interconnecting joint and pivotally coupled to a forward wing truss joint; a left diagonal linking member pivotally coupled to an aft-left fuselage truss joint and pivotally coupled to a forward-left wing truss joint; and a right diagonal linking member pivotally coupled to an aft-right fuselage truss joint and pivotally coupled to a forward-right wing truss joint.

[0006] In some embodiments, the techniques described herein relate to a wing-fuselage truss joint assembly for an aircraft including a forward-left joining member, an aft-left joining member, a forward-right joining member, and an aft-right joining member configured to join a wing to a fuselage. Each of the joining members includes a wing truss joint mechanically coupled to a spar of the wing; a fuselage truss joint mechanically coupled to a bulkhead of the fuselage; and an upright member pivotally coupled to the wing truss joint at a first end and pivotally coupled to the fuselage truss joint at a second end; an interconnecting joint mechanically coupled to a forward bulkhead between the forward-left joining member and the forward-right joining member; a first interconnecting member and a second interconnecting member wherein the first interconnecting member is pivotally coupled to the interconnecting joint and pivotally coupled to a forward-right wing truss joint and the second interconnecting member is pivotally coupled to the interconnecting joint and pivotally coupled to a forward-left wing truss joint; a right linking member pivotally coupled to an aft-right wing truss joint and pivotally coupled to the forward-right wing truss joint; and a left linking member pivotally coupled to an aft-left wing truss joint and pivotally coupled to the forward-left wing truss joint; a left diagonal linking member pivotally coupled to an aft-left fuselage truss joint and pivotally coupled to the forward-left wing truss joint; and a right diagonal linking member pivotally coupled to an aft-right fuselage truss joint and pivotally coupled to the forward-right wing truss joint.

[0007] In some embodiments, the techniques described herein relate to a wing-fuselage truss joint assembly for an aircraft including a forward-left joining member, an aft-left joining member, a forward-right joining member, and an aft-right joining member configured to join a wing to a fuselage. Each of the joining members includes a wing truss joint mechanically coupled to a spar of the wing; a fuselage truss joint mechanically coupled to a bulkhead of the fuselage; and an upright member pivotally coupled to the wing truss joint at a first end and pivotally coupled to the fuselage truss joint at a second end; an interconnecting joint mechanically coupled to a forward bulkhead between the forward-left joining member and the forward-right joining member; an interconnecting member pivotally coupled to the interconnecting joint and pivotally coupled to a forward wing truss joint; a left diagonal linking member pivotally coupled to an aft-left fuselage truss joint and pivotally coupled to a forward-left wing truss joint; and a right diagonal linking member pivotally coupled to an aft-right fuselage truss joint and pivotally coupled to a forward-right wing truss joint; an aft truss member pivotally coupled from an aft bulkhead to an aft wing truss joint.

## Description

BRIEF DESCRIPTION OF SEVERAL VIEWS OF THE DRAWINGS

[0008] Illustrative embodiments are described in detail below with reference to the attached drawing figures, which are incorporated by reference herein and wherein:

[0009] FIG. **1** is a perspective view showing a wing mounted to an aircraft fuselage via a wing-fuselage truss joint, in an embodiment;

[0010] FIG. **2**A is a front perspective view showing a front joining member of the wing-fuselage truss joint of FIG. **1**;

[0011] FIG. **2**B is a rear perspective view showing the front joining member of the wing-fuselage truss joint of FIG. **1**;

[0012] FIG. **2**C is a perspective view showing an interconnecting joint of the wing-fuselage truss joint of FIG. **1**;

[0013] FIG. **2**D is a front perspective view showing a rear joining member of the wing-fuselage truss joint of FIG. **1**;

[0014] FIG. **2**E is a rear perspective view showing the rear joining member of the wing-fuselage truss joint of FIG. **1**;

[0015] FIG. **2**F is a front perspective view of the front joining member of the wing-fuselage truss joint of FIG. **1** installed onto a forward bulkhead;

[0016] FIG. **3** is a side view showing the wing-fuselage truss joint of FIG. **1**;

[0017] FIG. **4**A is a perspective view of a first configuration of the wing-fuselage truss joint in an embodiment;

[0018] FIG. **4**B is a perspective view of the first configuration of the wing-fuselage truss joint of FIG. **4**A mounted to an aircraft structure;

[0019] FIG. **5**A is a perspective view showing a second configuration of the wing-fuselage truss joint mounted to an aircraft structure in an embodiment;

[0020] FIG. **5**B is a perspective view opposite that of FIG. **5**A showing the second configuration of the wing-fuselage truss joint mounted to an aircraft structure;

[0021] FIG. **6**A is a perspective view showing a third configuration of the wing-fuselage truss joint mounted to an aircraft structure in an embodiment; and

[0022] FIG. **6**B is a perspective view opposite that of FIG. **6**A showing the third configuration of the wing-fuselage truss joint mounted to an aircraft structure.

[0023] The drawing figures do not limit the invention to the specific embodiments disclosed and described herein. The drawings are not necessarily to scale, emphasis instead being placed upon clearly illustrating the principles of the invention.

DETAILED DESCRIPTION

[0024] The following detailed description references the accompanying drawings that illustrate specific embodiments in which the invention can be practiced. The embodiments are intended to describe aspects of the invention in sufficient detail to enable those skilled in the art to practice the invention. Other embodiments can be utilized and changes can be made without departing from the scope of the invention. The following detailed description is, therefore, not to be taken in a limiting sense. The scope of the invention is defined only by the appended claims, along with the full scope of equivalents to which such claims are entitled.

[0025] In this description, references to “one embodiment,” “an embodiment,” or “embodiments” mean that the feature or features being referred to are included in at least one embodiment of the technology. Separate references to “one embodiment,” “an embodiment,” or “embodiments” in this description do not necessarily refer to the same embodiment and are also not mutually exclusive unless so stated and/or except as will be readily apparent to those skilled in the art from the description. For example, a feature, structure, act, etc. described in one embodiment may also be included in other embodiments but is not necessarily included. Thus, the technology can include a variety of combinations and/or integrations of the embodiments described herein.

[0026] Connectors with high strength used for joining a wing and a fuselage together on an aircraft are essential for an aircraft structure to withstand aerodynamic and other types of loading. In current arrangements, a wing-fuselage joint may include a bolted joint connection which has high strength, but may have flexibility restrictions, and may require highly complex tooling for manufacture, increasing the cost and time of production. Additionally, specialized analysis is required for current wing-fuselage connections due to the presence of stress concentrations. The connectors must also be able to be rapidly assembled or disassembled to enable removal of the wing for facile transportation of the aircraft. A wing-fuselage joint is needed which may be assembled quickly while also being able to withstand various types of loads such as aerodynamic, ground reaction, vibration, shock, and thermal.

[0027] Embodiments disclosed herein provide an arrangement for connecting the wing and fuselage structures of an aircraft that incorporates truss members. The wing-fuselage truss joint in embodiments comprises a truss linkage including multiple linkage elements arranged in different orientations and planes. The wing-fuselage truss joint is designed to carry force loads in different directions while the truss linkage elements create a strong and flexible joint capable of withstanding numerous different types of loading such as aerodynamic, ground reaction, vibration, shock, and thermal types of loads. The wing-fuselage truss joint transfers shear forces experienced by an aircraft's wings in numerous directions with respect to the fuselage while substantially preventing the bending moment from being transferred to the fuselage structure. The substantial reduction or elimination of the bending moment to the fuselage allows for the fuselage size, complexity, cost, and manufacture time, to be reduced.

[0028] FIG. **1** shows a wing-fuselage truss joint assembly **100** mounted onto an aircraft structure in an embodiment. The wing-fuselage truss joint assembly **100** is configured for mounting a wing **104** to a fuselage structure **102**. The fuselage structure **102** for example is a pair of adjacent girderss or fuselage rib-like members of a frame and includes the forward bulkhead **103** and the aft bulkhead **106**. The wing **104** includes forward spar **107** and aft spar **109** which provide structural support for the wing **104** (FIG. **4**B). The bulkheads **103** and **106** of the fuselage structure **102** and the spars **107** and **109** of the wing **104** provide structural support for the fuselage and the wings of an aircraft respectively. The wing-fuselage truss joint assembly **100** is mounted to members of both the fuselage structure **102** and the wing **104** which mechanically couples the fuselage structure **102** and the wing **104** together. In embodiments, the wing **104** is configured as an upper wing mounted to a top side of the fuselage structure **102**.

[0029] The wing-fuselage truss joint assembly **100** comprises four joining members including a forward wing truss joint **124**, a forward fuselage truss joint **130**, an aft wing truss joint **136**, and an aft fuselage truss joint **140**. The forward wing truss joint **124** mechanically couples to the forward spar **107** of the wing **104**, the aft wing truss joint **136** mechanically couples to the aft spar **109**, the forward fuselage truss joint **130** mechanically couples to the forward bulkhead **103** and the aft fuselage truss joint **140** mechanically couples to the aft bulkhead **106**. An interconnecting joint **113** is mechanically coupled to the forward bulkhead **103**. Interconnecting members **112** are configured to span between the forward wing truss joints **124** and the interconnecting joint **113**.

[0030] In embodiments and with reference to FIG. **1**, the X direction is defined as a longitudinal direction aligned forward and aft along the length of the fuselage, the Y direction is a lateral direction that spans left and right across the aircraft in the main direction of the wings, and the Z direction is a vertical direction aligned up and down. The wing-fuselage truss joint assembly **100** is configured to reduce reaction forces in the X and Y directions from transferring to the fuselage structure **102** while substantially preventing moment forces and bending moments from being transferred to the fuselage structure **102**.

[0031] FIG. **2**A shows a close-up front perspective view of a forward joining member including the forward wing truss joint **124** and forward fuselage truss joint **130** shown in FIG. **1** on the forward left and forward right of the wing-fuselage truss joint assembly **100**. The forward wing truss joint **124** includes a base plate **126** including securing arrangements **116**. The securing arrangements **116** may be a nut and bolt arrangement and are configured on the base plate **126** to mount the forward wing truss joint **124** to a structural member of the wing **104**, such as the forward spar **107**. The base plate includes a flanged extension **128** which downwardly extends and curves to an end configured to pivotally receive an end of a truss link member **120** and an end of the diagonal truss link member **114**. The truss link member **120** and the forward wing truss joint **124** are coupled pivotally together such that each may pivot in the X-Z plane relative to the forward wing truss joint **124**. One securing arrangement **116** on the base plate **126** includes a pivotally mounted pivot bar **127** which is coupled to the interconnecting member **112** (FIG. **2**C). The interconnecting member **112** is pivotally mounted to the pivot bar **127** at one end and is pivotally mounted at its other end to the interconnecting joint **113** (FIG. **2**C). A vertical link **110** is configured to attach to the base plate **126** and extend downwards connecting to the forward fuselage truss joint **130**. The vertical link **110** is pivotally received by the forward fuselage truss joint **130** into a channel between two flanges **131** each extending away from a mounting base **132**. The flanges **131** include a securing element **133** extending laterally across the channel pivotally securing the vertical link **110** to the forward fuselage truss joint **130**. The mounting base **132** includes securing arrangements **118** which are each configured to secure the forward fuselage truss joint **130** to members of the fuselage structure **102**.

[0032] FIG. **2**B shows a close-up rear perspective view of the forward joining member of FIG. **2**A. The truss link member **120** includes a channeled end **125**. The channeled end **125** includes sidewalls spaced apart such that an end of the diagonal truss link member **114** may be pivotally secured in between the sidewalls of the channeled end **125**. The channeled end **125** of the truss link member **120** and the diagonal truss link member **114** are pivotally secured at the end of the flanged extension **128** of the forward wing truss joint **124**. The forward fuselage truss joint **130** is pivotally secured at a lower end of the vertical link **110** by the securing element **133** laterally extending between the two flanges **131** outcropped from the mounting base **132**. At each of the four corners of the square shaped mounting base **132**, the securing arrangements **118** are each inserted into a fastener hole in the mounting base **132**. The securing arrangement **118** may be a nut and bolt configured such that the bolt head is positioned on the upper surface of the mounting base **132** such that the shaft of the bolt extends downwards and through the fastener hole in the mounting base **132**. As each bolt shaft extends downwards one end of a fastener plate **129** is slid onto the bolt shaft with a nut being threadingly secured onto the upper end of the bolt shaft. The mounting base **132** includes two fastener plates **129** with one fastener plate **129** being inserted on one end of the forward-right securing arrangement **118** and on the other end being inserted onto the back-right securing arrangement **118**. Another fastener plate **129** is inserted onto the forward-left securing arrangement **118** and on the other end is inserted onto the back-left securing arrangement **118**. The fastener plates **129** span from the forward to the back edges of the mounting base **132** and are positioned along the left and right sides of the mounting base **132**. The fastener plates **129** are used to secure the forward fuselage truss joint **130** to the forward bulkhead **103** of the fuselage structure **102**. FIG. **2**F shows a perspective view of the forward fuselage truss joint **130** wherein the forward bulkhead **103** is sandwiched in between the fastener plates **129** and the mounting base **132** with the securing arrangements **118** passing through the forward bulkhead **103**. The securing arrangements **118** may be tightened to secure the forward bulkhead **103** in between the fastener plates **129** and mounting base **132**.

[0033] FIG. **2**C shows a close-up perspective view of the interconnecting joint **113** with the interconnecting members **112** connecting the interconnecting joint **113** to the forward joining members and more specifically the forward wing truss joints **124**. The interconnecting joint **113** includes a base plate **160** which includes securing arrangements **134** positioned on opposing sides which pivotally secure an end of each interconnecting member **112** connecting to the forward wing truss joints **124**. The interconnecting joint **113** includes a base **162** which having fasteners **135** configured to secure the interconnecting joint **113** to the fuselage structure **102**. The pivotal attachment of each of the interconnecting members **112** allows for the forward wing truss joint **124** to pivot relative to the fuselage structure **102** and the interconnecting joint **113**.

[0034] FIG. **2**D shows a close-up front perspective view of an aft joining member including the aft wing truss joint **136** and aft fuselage truss joint **140** as shown in FIG. **1** on the aft right and the aft left of the wing-fuselage truss joint assembly **100**. The aft wing truss joint **136** includes a base plate **138** including the securing arrangements **116**. The securing arrangements **116** are similar to those on the forward wing truss joint **124** and are configured on the base plate **138** to mount the aft wing truss joint **136** to a structural member such as the aft spar **109** of the wing **104**. The base plate **138** includes a pivotal connection **139** wherein the vertical link **110** and the truss link member **120** are pivotally coupled together and allow the truss link member **120** and the vertical link **110** to pivot in the X-Z plane independently of one another. The vertical link **110** extends downwards to the aft fuselage truss joint **140**. The aft fuselage truss joint **140** is configured with a channeled aperture **142** opening having a pivotal connection **144** to pivotally connect the vertical link **110**.

[0035] FIG. **2**E shows a rear perspective view of the aft joining member of FIG. **2**D. The aft fuselage truss joint **140** includes fastening bolts **148** and shear pins **149** used to secure the aft fuselage truss joint **140** to the aft bulkhead **106** (FIG. **3**). In embodiments, the fastening bolts **148** may be a nut, washer, and bolt arrangement. In embodiments when viewing FIGS. **2**D and **2**E together, the fastening bolts **148** are configured such that the fastening bolts **148** are positioned perpendicularly to the shear pins **149** allowing the aft fuselage truss joint **140** to be secured onto an edged side of the aft bulkhead **106** (as shown in FIG. **3**). In embodiments, the shear pins **149** are laminated and are configured to substantially carry the vertical load on the aft fuselage truss joint **140**. The aft fuselage truss joint **140** includes the pivotal connection **146** pivotally connecting the diagonal truss link member **114** to the aft fuselage truss joint **140**.

[0036] The interconnecting members **112** and the interconnecting joint **113** (FIG. **2**C) allows for forces experienced by the wings in the Y direction to be transferred to the fuselage structure **102**. The interconnecting member **112** is pivotally connected on one end to the interconnecting joint **113** and on the other end to the pivot bar **127**. The pivot bar **127** is pivotally connected to the forward wing truss joint **124** via securing arrangement **116**. The linkage created between the interconnecting joint **113**, interconnecting members **112**, the pivot bar **127**, and the forward wing truss joint **124** allow for each forward wing truss joint **124** to pivot in the Y-Z plane relative to the aircraft fuselage structure **102**. For instance, when a force is experienced by the wing **104**, the pivotal connection allows for the interconnecting members **112** to pivot in the Y-Z plane relative to the interconnecting joint **113** mounted to the fuselage structure **102** (FIG. **1**). Movement of the interconnecting members **112** in the Y-Z plane substantially allows for the forces experienced by the wing **104** in the Y direction to be transferred to the fuselage structure **102**.

[0037] FIG. **3** shows a side view of the wing-fuselage truss joint assembly **100**. The forward wing truss joint **124** and the aft wing truss joint **136** are connected to the forward spar **107** and the aft spar **109** of the wing **104** respectively, and the forward fuselage truss joint **130** and aft fuselage truss joint **140** are connected, respectively, to the forward bulkhead **103** and aft bulkhead **106** of the fuselage structure **102**. The truss link member **120** is pivotally connected at one end to the aft wing truss joint **136** at the pivotal connection **139** and at the other end to the the flanged extension **128** of the forward wing truss joint **124**. The diagonal truss link member **114** is pivotally connected at one end to the pivotal connection **146** on the aft fuselage truss joint **140** and at the other end to the flanged extension **128** of the forward wing truss joint **124**. The pivotal connection of the diagonal truss link member **114** between the aft fuselage truss joint **140** and the forward wing truss joint **124** limits the wing **104** from pivoting in the X-Z plane and limits the movement of the wing **104** in the X direction relative to the fuselage structure **102**. For instance, when a force is experienced by the wing **104**, the pivotal connection of the diagonal truss link member **114** limits the wing **104** from pivoting in the X-Z plane and limiting the movement of the wing **104** in the X direction (i.e., front and back). Limited movement of the diagonal truss link member **114** in the X-Z plane substantially allows for the forces experienced by the wing **104** in the X direction to be transferred to the fuselage structure **102**.

[0038] The wing-fuselage truss joint assembly **100** allows for the forces experienced by the wing **104** to be transferred to the fuselage structure **102** in the X, Y, and Z directions while limiting and substantially eliminating the bending moment from being transferred to the fuselage structure **102**. The interconnecting joint **113** is configured to allow the forward wing truss joint **124** to pivot in the Y-Z plane (see FIG. **2**C) while the truss link member **120** and the diagonal truss link member **114** (see FIG. **3**) allow the forward wing truss joint **124** and aft wing truss joint **136** to be substantially connected. The vertical link **110** and interconnecting members **112** each pivotally connected to the forward wing truss joint **124** and substantially prevent a moment force experienced by the wings **104** to be transferred to the fuselage structure **102**. The allowed pivotal movement of the interconnecting members **112** (Y-Z plane) and the diagonal truss link member **114** and truss link member **120** (X-Z plane) allow for substantially less of the forces experienced by the wing **104** in the X and Y directions to be transferred from the wing **104** to the fuselage structure **102**.

[0039] The wing-fuselage truss joint assembly **100** may be configured in numerous different ways to provide desired force transmission to the fuselage. FIG. **4**A shows a perspective view of the wing-fuselage truss joint assembly **100**. The wing-fuselage truss joint assembly **100** includes a forward wing truss joint **124** and a forward fuselage truss joint **130**, and an aft wing truss joint **136** and an aft fuselage truss joint **140**. The diagonal truss link member **114** connects a forward wing truss joint **124** with an aft fuselage truss joint **140** on each side, and truss link member **120** connects a forward wing truss joint **124** with an aft wing truss joint **136** on each side. The forward wing truss joints **124** are each connected with an interconnecting joint **113** via one of the interconnecting members **112**. The wing-fuselage truss joint assembly **100** is shown mounted to the fuselage structure **102** and the wing **104** in FIG. **4**B. The wing-fuselage truss joint assembly **100** of FIG. **4**A and FIG. **4**B allows for the forces experienced by the wing **104** in the X and Y direction to be substantially reduced and transferred to the fuselage structure **102** while the forces created from a bending moment in the wing **104** is minimally transferred and possibly eliminated from travelling to the fuselage structure **102**.

[0040] FIGS. **5**A and **5**B show a wing-fuselage truss joint assembly **200**, which is similar to the wing-fuselage truss joint assembly **100**. FIG. **5**A provides a front perspective view of the wing-fuselage truss joint assembly **200** and FIG. **5**B provides a rear perspective view of the wing-fuselage truss joint assembly **200**. The wing-fuselage truss joint assembly **200** is configured similarly to wing-fuselage truss joint assembly **100** of FIG. **4**A and FIG. **4**B, except with additional

truss members added. Specifically, the wing-fuselage truss joint assembly **200** further includes a pair of aft truss members **202** connected from the aft wing truss joint **136** directly to the aft bulkhead **106** of the fuselage structure **102** instead of using an interconnecting joint **113**. The aft truss members **202** are configured on each of the aft wing truss joints **136** and extend downwards at an angle to be mounted to the aft bulkhead **106** of the fuselage structure **102**. The wing-fuselage truss joint assembly **200** may be used when additional truss links are required for transmission of force loads in the Y direction. The aft truss members **202** are pivotally connected to each aft wing truss joint **136** and pivotally connected to the fuselage structure **102** (aft bulkhead **106**) such that forces in the Y direction are transferred to the fuselage structure **102** and rotational moments, such as bending moments, are not transferred to the fuselage structure **102**.
[0041] FIGS. **6**A and **6**B show a wing-fuselage truss joint assembly **300**, which is similar to the wing-fuselage truss joint assembly **100**. FIG. **6**A provides a front perspective view of the wing-fuselage truss joint assembly **300** and FIG. **6**B provides a rear perspective view of the wing-fuselage truss joint assembly **300**. The wing-fuselage truss joint assembly **300** is configured similarly to the first and second wing-fuselage truss joint assemblies **100** and **200** with some truss members eliminated. Specifically, in wing-fuselage truss joint assembly **300**, the truss link members **120**, a single interconnecting member **112**, and a single rear truss member **202** are eliminated. The wing-fuselage truss joint assembly **300** may be used for transmission of force loads in all directions while limiting the amount of bending forces transferred to the fuselage structure **102** and being able to provide strength for rotational loads or moments around the Z-axis. In comparison, the wing-fuselage truss joint assembly **200** may allow more bending forces to transfer to the fuselage structure **102** than the wing-fuselage truss joint assembly **300**.
[0042] Features described above as well as those claimed below may be combined in various ways without departing from the scope hereof. The following clauses illustrate some possible, non-limiting combinations:
[0043] Clause 1. A wing-fuselage truss joint assembly for an aircraft comprising: a forward-left joining member, an aft-left joining member, a forward-right joining member, and an aft-right joining member configured to join a wing to a fuselage, wherein each of the joining members comprises: a wing truss joint mechanically coupled to a spar of the wing; a fuselage truss joint mechanically coupled to a bulkhead of the fuselage; and an upright member pivotally coupled to the wing truss joint at a first end and pivotally coupled to the fuselage truss joint at a second end; an interconnecting joint mechanically coupled to a forward bulkhead between the forward-left joining member and the forward-right joining member; at least one interconnecting member pivotally coupled to the interconnecting joint and pivotally coupled to a forward wing truss joint; a left diagonal linking member pivotally coupled to an aft-left fuselage truss joint and pivotally coupled to a forward-left wing truss joint; and a right diagonal linking member pivotally coupled to an aft-right fuselage truss joint and pivotally coupled to a forward-right wing truss joint.
[0044] Clause 2. The wing-fuselage truss joint assembly of clause 1 wherein a right truss link member pivotally connects from the aft-right joining member to the forward-right joining member and a left truss link member pivotally connects from the aft-left joining member to the forward-left joining member.
[0045] Clause 3. The wing-fuselage truss joint assembly of clauses 1-2 wherein the right truss link member pivotally connects from an aft-right wing truss joint to the forward-right wing truss joint and the left truss link member pivotally connects from an aft-left wing truss joint to the forward-left wing truss joint.
[0046] Clause 4. The wing-fuselage truss joint assembly of clauses 1-3 wherein a first aft truss member connects from an aft bulkhead to an aft-right wing truss joint and a second aft truss member connects from the aft bulkhead to an aft-left wing truss joint.
[0047] Clause 5. The wing-fuselage truss joint assembly of clauses 1-4 wherein the forward wing truss joint comprises a pivot bar configured to pivotally connect with the at least one

interconnecting member.

[0048] Clause 6. The wing-fuselage truss joint assembly of clauses 1-5 wherein the pivot bar is configured to pivot in a first direction, the first direction being a lateral direction aligned with the wing.

[0049] Clause 7. The wing fuselage joint assembly of clauses 1-6 wherein the left diagonal linking member and the right diagonal linking member pivot in a second direction, the second direction being a longitudinal direction aligned with the fuselage.

[0050] Clause 8. The wing-fuselage truss joint assembly of clauses 1-7 wherein the right truss link member and the left truss link member are each configured to pivot in the second direction.

[0051] Clause 9. The wing-fuselage truss joint assembly of clauses 1-8, wherein the joining members, the at least one interconnecting member, the left and right diagonal linking members, and the left and right truss link members are configured to substantially reduce forces experienced by the wing from being transferred to a fuselage structure.

[0052] Clause 10. A wing-fuselage truss joint assembly for an aircraft comprising: a forward-left joining member, an aft-left joining member, a forward-right joining member, and an aft-right joining member configured to join a wing to a fuselage, wherein each of the joining members comprises: a wing truss joint mechanically coupled to a spar of the wing; a fuselage truss joint mechanically coupled to a bulkhead of the fuselage; and an upright member pivotally coupled to the wing truss joint at a first end and pivotally coupled to the fuselage truss joint at a second end; an interconnecting joint mechanically coupled to a forward bulkhead between the forward-left joining member and the forward-right joining member; a first interconnecting member and a second interconnecting member wherein the first interconnecting member is pivotally coupled to the interconnecting joint and pivotally coupled to a forward-right wing truss joint and the second interconnecting member is pivotally coupled to the interconnecting joint and pivotally coupled to a forward-left wing truss joint; a right linking member pivotally coupled to an aft-right wing truss joint and pivotally coupled to the forward-right wing truss joint; and a left linking member pivotally coupled to an aft-left wing truss joint and pivotally coupled to the forward-left wing truss joint; a left diagonal linking member pivotally coupled to an aft-left fuselage truss joint and pivotally coupled to the forward-left wing truss joint; and a right diagonal linking member pivotally coupled to an aft-right fuselage truss joint and pivotally coupled to the forward-right wing truss joint.

[0053] Clause 11. The wing-fuselage truss joint assembly of clause 10 including a first aft truss member connecting from an aft bulkhead to the aft-right wing truss joint and a second aft truss member connecting from the aft bulkhead to the aft-left wing truss joint.

[0054] Clause 12. The wing-fuselage truss joint assembly of clause 10 or 11 wherein the aft-right wing truss joint includes a first pivot bar and the aft-left wing truss joint includes a second pivot bar wherein the first pivot bar is configured to pivotally receive an end of the first aft truss member and the second pivot bar is configured to pivotally receive an end of the second aft truss member.

[0055] Clause 13. The wing-fuselage truss joint assembly of clauses 10-12 wherein the first pivot bar and the second pivot bar pivot in a first direction, the first direction being a lateral direction aligned with the wing.

[0056] Clause 14. The wing-fuselage truss joint assembly of clauses 10-13 wherein the fuselage truss joint is configured such that the upright member is configured to pivot in the first direction at its second end.

[0057] Clause 15. The wing-fuselage truss joint assembly of clauses 10-14 wherein the wing truss joint is configured such that the upright member is configured to pivot in a second direction at its first end, the second direction being a longitudinal direction aligned with the fuselage.

[0058] Clause 16. A wing-fuselage truss joint assembly for an aircraft comprising: a forward-left joining member, an aft-left joining member, a forward-right joining member, and an aft-right joining member configured to join a wing to a fuselage, wherein each of the joining members

comprises: a wing truss joint mechanically coupled to a spar of the wing; a fuselage truss joint mechanically coupled to a bulkhead of the fuselage; and an upright member pivotally coupled to the wing truss joint at a first end and pivotally coupled to the fuselage truss joint at a second end; an interconnecting joint mechanically coupled to a forward bulkhead between the forward-left joining member and the forward-right joining member; an interconnecting member pivotally coupled to the interconnecting joint and pivotally coupled to a forward wing truss joint; a left diagonal linking member pivotally coupled to an aft-left fuselage truss joint and pivotally coupled to a forward-left wing truss joint; and a right diagonal linking member pivotally coupled to an aft-right fuselage truss joint and pivotally coupled to a forward-right wing truss joint; an aft truss member pivotally coupled from an aft bulkhead to an aft wing truss joint.
[0059] Clause 17. The wing-fuselage joint assembly of clause 16 wherein the interconnecting member is pivotally coupled to the forward-left wing truss joint and the aft truss member is pivotally coupled to the aft-right wing truss joint.
[0060] Clause 18. The wing-fuselage truss joint assembly of clause 16 or 17 wherein the interconnecting member is pivotally coupled to the forward-right wing truss joint and the aft truss member is pivotally coupled to the aft-left wing truss joint.
[0061] Clause 19. The wing fuselage truss joint assembly of clauses 16-18 wherein the forward-left wing truss joint and the forward-right wing truss joint each include a flanged extension configured to pivot in a second direction, the second direction being a longitudinal direction aligned with the fuselage.
[0062] Clause 20. The wing-fuselage joint of clauses 16-19 wherein the left diagonal linking member is pivotally coupled to the flanged extension and the right diagonal linking member is pivotally coupled to the flanged extension.
[0063] Many different arrangements of the various components depicted, as well as components not shown, are possible without departing from the spirit and scope of what is claimed herein. Embodiments have been described with the intent to be illustrative rather than restrictive. Alternative embodiments will become apparent to those skilled in the art that do not depart from what is disclosed. A skilled artisan may develop alternative means of implementing the aforementioned improvements without departing from what is claimed.
[0064] It will be understood that certain features and subcombinations are of utility and may be employed without reference to other features and subcombinations and are contemplated within the scope of the claims. Not all steps listed in the various figures need be carried out in the specific order described.

## Claims

**1**. A wing-fuselage truss joint assembly for an aircraft comprising: a forward-left joining member, an aft-left joining member, a forward-right joining member, and an aft-right joining member configured to join a wing to a fuselage, wherein each of the joining members comprises: a wing truss joint mechanically coupled to a spar of the wing; a fuselage truss joint mechanically coupled to a bulkhead of the fuselage; and an upright member pivotally coupled to the wing truss joint at a first end and pivotally coupled to the fuselage truss joint at a second end; an interconnecting joint mechanically coupled to a forward bulkhead between the forward-left joining member and the forward-right joining member; at least one interconnecting member pivotally coupled to the interconnecting joint and pivotally coupled to a forward wing truss joint; a left diagonal linking member pivotally coupled to an aft-left fuselage truss joint and pivotally coupled to a forward-left wing truss joint; and a right diagonal linking member pivotally coupled to an aft-right fuselage truss joint and pivotally coupled to a forward-right wing truss joint.
**2**. The wing-fuselage truss joint assembly of claim 1 wherein a right truss link member pivotally connects from the aft-right joining member to the forward-right joining member and a left truss

link member pivotally connects from the aft-left joining member to the forward-left joining member.

**3**. The wing-fuselage truss joint assembly of claim 2 wherein the right truss link member pivotally connects from an aft-right wing truss joint to the forward-right wing truss joint and the left truss link member pivotally connects from an aft-left wing truss joint to the forward-left wing truss joint.

**4**. The wing-fuselage truss joint assembly of claim 1 wherein a first aft truss member connects from an aft bulkhead to an aft-right wing truss joint and a second aft truss member connects from the aft bulkhead to an aft-left wing truss joint.

**5**. The wing-fuselage truss joint assembly of claim 1 wherein the forward wing truss joint comprises a pivot bar configured to pivotally connect with the at least one interconnecting member.

**6**. The wing-fuselage truss joint assembly of claim 1 wherein the pivot bar is configured to pivot in a first direction, the first direction being a lateral direction aligned with the wing.

**7**. The wing fuselage joint assembly of claim 1 wherein the left diagonal linking member and the right diagonal linking member pivot in a second direction, the second direction being a longitudinal direction aligned with the fuselage.

**8**. The wing-fuselage truss joint assembly of claim 2 wherein the right truss link member and the left truss link member are each configured to pivot in the second direction.

**9**. The wing-fuselage truss joint assembly of claim 2, wherein the joining members, the at least one interconnecting member, the left and right diagonal linking members, and the left and right truss link members are configured to substantially reduce forces experienced by the wing from being transferred to a fuselage structure.

**10**. A wing-fuselage truss joint assembly for an aircraft comprising: a forward-left joining member, an aft-left joining member, a forward-right joining member, and an aft-right joining member configured to join a wing to a fuselage, wherein each of the joining members comprises: a wing truss joint mechanically coupled to a spar of the wing; a fuselage truss joint mechanically coupled to a bulkhead of the fuselage; and an upright member pivotally coupled to the wing truss joint at a first end and pivotally coupled to the fuselage truss joint at a second end; an interconnecting joint mechanically coupled to a forward bulkhead between the forward-left joining member and the forward-right joining member; a first interconnecting member and a second interconnecting member wherein the first interconnecting member is pivotally coupled to the interconnecting joint and pivotally coupled to a forward-right wing truss joint and the second interconnecting member is pivotally coupled to the interconnecting joint and pivotally coupled to a forward-left wing truss joint; a right linking member pivotally coupled to an aft-right wing truss joint and pivotally coupled to the forward-right wing truss joint; and a left linking member pivotally coupled to an aft-left wing truss joint and pivotally coupled to the forward-left wing truss joint; a left diagonal linking member pivotally coupled to an aft-left fuselage truss joint and pivotally coupled to the forward-left wing truss joint; and a right diagonal linking member pivotally coupled to an aft-right fuselage truss joint and pivotally coupled to the forward-right wing truss joint.

**11**. The wing-fuselage truss joint assembly of claim 10 including a first aft truss member connecting from an aft bulkhead to the aft-right wing truss joint and a second aft truss member connecting from the aft bulkhead to the aft-left wing truss joint.

**12**. The wing-fuselage truss joint assembly of claim 11 wherein the aft-right wing truss joint includes a first pivot bar and the aft-left wing truss joint includes a second pivot bar wherein the first pivot bar is configured to pivotally receive an end of the first aft truss member and the second pivot bar is configured to pivotally receive an end of the second aft truss member.

**13**. The wing-fuselage truss joint assembly of claim 12 wherein the first pivot bar and the second pivot bar pivot in a first direction, the first direction being a lateral direction aligned with the wing.

**14**. The wing-fuselage truss joint assembly of claim 10 wherein the fuselage truss joint is configured such that the upright member is configured to pivot in the first direction at its second end.

**15**. The wing-fuselage truss joint assembly of claim 10 wherein the wing truss joint is configured such that the upright member is configured to pivot in a second direction at its first end, the second direction being a longitudinal direction aligned with the fuselage.
**16**. A wing-fuselage truss joint assembly for an aircraft comprising: a forward-left joining member, an aft-left joining member, a forward-right joining member, and an aft-right joining member configured to join a wing to a fuselage, wherein each of the joining members comprises: a wing truss joint mechanically coupled to a spar of the wing; a fuselage truss joint mechanically coupled to a bulkhead of the fuselage; and an upright member pivotally coupled to the wing truss joint at a first end and pivotally coupled to the fuselage truss joint at a second end; an interconnecting joint mechanically coupled to a forward bulkhead between the forward-left joining member and the forward-right joining member; an interconnecting member pivotally coupled to the interconnecting joint and pivotally coupled to a forward wing truss joint; a left diagonal linking member pivotally coupled to an aft-left fuselage truss joint and pivotally coupled to a forward-left wing truss joint; and a right diagonal linking member pivotally coupled to an aft-right fuselage truss joint and pivotally coupled to a forward-right wing truss joint; an aft truss member pivotally coupled from an aft bulkhead to an aft wing truss joint.
**17**. The wing-fuselage joint assembly of claim 16 wherein the interconnecting member is pivotally coupled to the forward-left wing truss joint and the aft truss member is pivotally coupled to the aft-right wing truss joint.
**18**. The wing-fuselage truss joint assembly of claim 16 wherein the interconnecting member is pivotally coupled to the forward-right wing truss joint and the aft truss member is pivotally coupled to the aft-left wing truss joint.
**19**. The wing fuselage truss joint assembly of claim 16 wherein the forward-left wing truss joint and the forward-right wing truss joint each include a flanged extension configured to pivot in a second direction, the second direction being a longitudinal direction aligned with the fuselage.
**20**. The wing-fuselage joint of claim 19 wherein the left diagonal linking member is pivotally coupled to the flanged extension and the right diagonal linking member is pivotally coupled to the flanged extension.

---